

Pranil Parajuli

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Profile

Computer Engineer graduate with hands-on industry experience in deep learning, computer vision, and generative AI.

Education

Bachelor of Computer Engineering

Thapathali Campus, Thapathali, Bagmati Province

Graduated: 2082 (Spring 2026)

Aggregate: 80.84%

Relevant Coursework: Machine Learning, Computer Vision, Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks, Digital Signal Processing

+2 / SLC

Fluorescent Secondary School, Kathmandu

Graduated: 2022 (2077 BS)

GPA: 3.72

SEE

Fluorescent Secondary School, Kathmandu

Graduated: 2020 (2075 BS)

GPA: 3.80

Research Experience

Kathmandu Valley Satellite Image Segmentation

[\[GitHub\]](#) 2025

Python · PyTorch · SegFormer-B0 · Hugging Face · Albumentations

- Created a **custom satellite dataset** of ~400 tiles (512×512) of Kathmandu Valley, manually annotated in CVAT across 7 land-use classes for urban planning and GIS applications.
- Fine-tuned **SegFormer-B0** (MiT encoder + MLP decoder) with AdamW, weighted cross-entropy for class imbalance, mixed-precision training, and **3-fold cross-validation** over 300 epochs.
- Published the fine-tuned model to **Hugging Face Hub** for public inference; monitored per-class **mIoU** across folds with stable gradient norms confirming effective convergence.

ControlNet + Stable Diffusion, Smart City Generation

[\[GitHub\]](#) 2025

Python · Stable Diffusion · ControlNet · Diffusers · PyTorch

- Built a generative pipeline combining **ControlNet with Stable Diffusion** to synthesise realistic smart-city satellite images conditioned on segmentation masks and text prompts.
- Engineered configurable inference supporting DPM++2M / Euler schedulers, LoRA weights, 2× upscaling, and batch processing; added a **low-VRAM mode** (CPU offloading) for 4GB GPUs.
- Implemented a spatial coverage analysis module and delivered full CLI documentation with usage examples.

Experience

AI/ML Engineer

Acaiberry Technologies, Lalitpur

November 2025 – March 2026

- Developed a real-time **Face Recognition system** using MTCNN and FaceNet (PyTorch, OpenCV), achieving reliable identity verification across varied lighting conditions.
- Implemented **Skin Analysis and Wrinkle Detection** using Hessian-based feature extraction to classify skin conditions and detect wrinkle patterns from facial images.
- Contributed to myabhiyan.com, a live campaign and social impact platform built by Acaiberry Technologies.
- Conducted **QSAR (Quantitative Structure Activity Relationship) research**, applying ML regression models to predict molecular biological activity from chemical descriptors.

Technical Skills

- Languages:** Python, Rust, C++, C, Golang, JavaScript, Assembly
- ML & AI:** Deep Learning, Computer Vision, LLMs, Generative AI (Stable Diffusion, ControlNet), Semantic Segmentation, Retrieval-Augmented Generation (RAG)

- **Frameworks & Libraries:** PyTorch, OpenCV, Hugging Face Transformers, Diffusers, Flask, LangChain
- **Web Development:** React (TypeScript), Node.js, HTML, CSS
- **Databases:** PostgreSQL, MySQL
- **Tools & Platforms:** Git, Linux, SFML

Projects

Retrieval-Augmented Generation (RAG) Pipeline

[\[GitHub\]](#) 2025

Python · PyTorch · Hugging Face Transformers · LangChain · Ollama (Qwen2.5)

- Built a complete RAG system from scratch: token-aware **chunking with overlap** (BAAI/bge-small-en-v1.5 tokenizer, configurable chunk size) and paragraph-level splitting to preserve semantic boundaries.
- Generated dense **sentence embeddings** via mean-pooled hidden states from a Hugging Face transformer model, then retrieved top-*k* chunks using **cosine similarity** over a JSON-backed vector store.
- Integrated **Ollama (Qwen2.5)** as the local LLM backend through LangChain prompt chaining, grounding responses strictly in retrieved context for hallucination reduction.

Interactive Chatbot for Mental Well-being

[\[GitHub\]](#) 2024

Python · PyTorch · Hugging Face · React (TypeScript) · Flask · PostgreSQL

- Fine-tuned **GPT-2 (124M parameters)** on 45K rows of mental health conversational data to generate empathetic, context-aware responses; evaluated model quality using **ROUGE** and **BLEU** metrics.
- Integrated the fine-tuned model into a full-stack application with real-time chat, **JWT-based user authentication**, and a feedback/rating system for continuous model improvement.
- Designed a PostgreSQL-backed analytics layer to track usage patterns and surface response quality insights.

Face Recognition Attendance System

[\[GitHub\]](#) 2025

Python · PyTorch · MTCNN · FaceNet · OpenCV · PostgreSQL

- Built a real-time attendance system using a **3-phase pipeline**: MTCNN detection, 68-point landmark alignment for pose normalisation, and **FaceNet (InceptionResnetV1, VGGFace2)** generating 512-d embeddings.
- Achieved robust identity verification via dual similarity matching (**Euclidean** ≤ 1.1 and **Cosine similarity** ≥ 0.6) with CUDA GPU acceleration.
- Persisted student records and face encodings in **PostgreSQL**, enabling scalable deployment for institutions and access control systems.

Handwritten Digit Recognition Neural Network

[\[GitHub\]](#) 2025

Python · NumPy · Matplotlib · MNIST (Kaggle)

- Implemented a **2-layer neural network from scratch** in NumPy ($784 \rightarrow 10 \rightarrow 10$) with ReLU hidden activation and Softmax output, trained on the MNIST digit dataset.
- Derived and coded **forward and backward propagation** manually, including He initialisation, one-hot encoding, and gradient descent weight updates, no ML framework used.
- Achieved reliable digit classification with per-sample prediction visualisation; normalised pixel inputs to $[0, 1]$ and validated on a held-out 1,000-sample dev set.

Demon War, 2D Action Game

[\[GitHub\]](#) 2024

Rust

- Architected a complete 2D game engine in **Rust** from scratch, covering game loop, collision detection, and sprite management.
- Designed a player ability system (Unbreakable Shield, Teleport, Ultimate) with dynamic **weapon switching** (Bow, Knife, Axe).
- Implemented distinct **enemy AI behaviours**: Bat (melee strike), Dinosaur (fast horizontal charge), Spider (web immobilisation).

Physics-Based Golf Game

[\[GitHub\]](#) 2024

C++ · SFML

- Developed a drag-to-aim **physics engine** for ball trajectory incorporating water, sand, and wind obstacles using SFML.

Sorting Algorithm Visualizer

[\[GitHub\]](#) 2024

C++ · SFML

- Rendered step-by-step animations of **Bubble Sort**, **Merge Sort (iterative & recursive)**, and **Quick Sort** to aid algorithmic understanding.

Finance Tracker

[\[GitHub\]](#) 2024

React (TypeScript) · PostgreSQL

- Built a personal finance dashboard tracking monthly income, expenses, and savings stored in **PostgreSQL**.
- Integrated **dynamic pie charts** for real-time spending pattern visualisation and trend analysis.

More projects at github.com/praniil